

Reduction of Quadratic Form to Canonical Form

Problem 14 - Complete Step-by-Step Solution

Problem 14:

Reduce the quadratic form

$$Q = x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_3x_1$$

to canonical form.

(10 marks)

Solution:

Step 1: Form the Symmetric Matrix A

Write the quadratic form in matrix notation:

$$Q = \mathbf{x}^T \mathbf{A} \mathbf{x}$$

where $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Identify coefficients from the quadratic form:

$$Q = x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_3x_1$$

- Coefficient of x_1^2 : $a_{11} = 1$
- Coefficient of x_2^2 : $a_{22} = 5$
- Coefficient of x_3^2 : $a_{33} = 1$
- Coefficient of x_1x_2 : Split as $a_{12} = a_{21} = \frac{2}{2} = 1$
- Coefficient of x_2x_3 : Split as $a_{23} = a_{32} = \frac{2}{2} = 1$

- Coefficient of x_3x_1 : Split as $a_{13} = a_{31} = \frac{6}{2} = 3$

Symmetric Matrix:

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Verification: Expand $x^T Ax$ to confirm:

$$\begin{aligned} & \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_1x_3 \end{aligned}$$

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Step 2: Find the Characteristic Equation and Eigenvalues

The characteristic equation is:

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$

Expand the determinant using first row:

$$(1-\lambda) \begin{vmatrix} 5-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 3 & 1-\lambda \end{vmatrix} + 3 \begin{vmatrix} 1 & 5-\lambda \\ 3 & 1 \end{vmatrix} = 0$$

Calculate each 2×2 determinant:

First determinant:

$$\begin{vmatrix} 5-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = (5-\lambda)(1-\lambda) - 1$$

$$= 5 - 5\lambda - \lambda + \lambda^2 - 1 = \lambda^2 - 6\lambda + 4$$

Second determinant:

$$\begin{vmatrix} 1 & 1 \\ 3 & 1-\lambda \end{vmatrix} = 1(1-\lambda) - 3 = 1 - \lambda - 3 = -2 - \lambda$$

Third determinant:

$$\begin{vmatrix} 1 & 5-\lambda \\ 3 & 1 \end{vmatrix} = 1 - 3(5-\lambda) = 1 - 15 + 3\lambda = 3\lambda - 14$$

Combine all terms:

$$(1-\lambda)(\lambda^2 - 6\lambda + 4) - 1(-2 - \lambda) + 3(3\lambda - 14) = 0$$

$$(1-\lambda)(\lambda^2 - 6\lambda + 4) + 2 + \lambda + 9\lambda - 42 = 0$$

Expand $(1-\lambda)(\lambda^2 - 6\lambda + 4)$:

$$\lambda^2 - 6\lambda + 4 - \lambda^3 + 6\lambda^2 - 4\lambda = 0$$

$$-\lambda^3 + 7\lambda^2 - 10\lambda + 4 + 2 + \lambda + 9\lambda - 42 = 0$$

$$-\lambda^3 + 7\lambda^2 + 0\lambda - 36 = 0$$

$$\lambda^3 - 7\lambda^2 + 36 = 0$$

Solve by trial and error or factoring:

Try $\lambda = -2$: $(-2)^3 - 7(-2)^2 + 36 = -8 - 28 + 36 = 0 \checkmark$

So $(\lambda + 2)$ is a factor.

Perform polynomial division:

$$\lambda^3 - 7\lambda^2 + 36 = (\lambda + 2)(\lambda^2 - 9\lambda + 18)$$

Factor the quadratic:

$$\lambda^2 - 9\lambda + 18 = (\lambda - 3)(\lambda - 6)$$

Eigenvalues:

$$\lambda_1 = -2, \quad \lambda_2 = 3, \quad \lambda_3 = 6$$

Step 3: Find Eigenvectors

For $\lambda_1 = -2$:

Solve $(A - (-2)I)v = 0$ or $(A + 2I)v = 0$:

$$A + 2I = \begin{bmatrix} 3 & 1 & 3 \\ 1 & 7 & 1 \\ 3 & 1 & 3 \end{bmatrix}$$

Set up the system:

$$\begin{bmatrix} 3 & 1 & 3 \\ 1 & 7 & 1 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Row reduce:

$R_1 \leftrightarrow R_2$ (swap for convenience):

$$\begin{bmatrix} 1 & 7 & 1 \\ 3 & 1 & 3 \\ 3 & 1 & 3 \end{bmatrix}$$

$R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 7 & 1 \\ 0 & -20 & 0 \\ 0 & -20 & 0 \end{bmatrix}$$

From R_2 : $-20v_2 = 0 \Rightarrow v_2 = 0$

From R_1 : $v_1 + 7(0) + v_3 = 0 \Rightarrow v_1 = -v_3$

Choose $v_3 = 1$:

Then $v_1 = -1$ and $v_2 = 0$

First eigenvector: $v_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

Normalize:

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{1+0+1}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 3$:

Solve $(A - 3I)v = 0$:

$$A - 3I = \begin{bmatrix} -2 & 1 & 3 \\ 1 & 2 & 1 \\ 3 & 1 & -2 \end{bmatrix}$$

Row reduce:

$R_1 \leftrightarrow R_2$:

$$\begin{bmatrix} 1 & 2 & 1 \\ -2 & 1 & 3 \\ 3 & 1 & -2 \end{bmatrix}$$

$R_2 \rightarrow R_2 + 2R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 5 & 5 \\ 0 & -5 & -5 \end{bmatrix}$$

$R_2 \rightarrow R_2/5$:

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -5 & -5 \end{bmatrix}$$

From R_2 : $v_2 + v_3 = 0 \Rightarrow v_2 = -v_3$

From R_1 : $v_1 + 2v_2 + v_3 = 0 \Rightarrow v_1 + 2(-v_3) + v_3 = 0 \Rightarrow v_1 = v_3$

Choose $v_3 = 1$:

Then $v_1 = 1$ and $v_2 = -1$

Second eigenvector: $v_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

Normalize:

$$u_2 = \frac{1}{\sqrt{1+1+1}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

For $\lambda_3 = 6$:

Solve $(A - 6I)v = 0$:

$$A - 6I = \begin{bmatrix} -5 & 1 & 3 \\ 1 & -1 & 1 \\ 3 & 1 & -5 \end{bmatrix}$$

Row reduce:

$R_1 \leftrightarrow R_2$:

$$\begin{bmatrix} 1 & -1 & 1 \\ -5 & 1 & 3 \\ 3 & 1 & -5 \end{bmatrix}$$

$R_2 \rightarrow R_2 + 5R_1$ and $R_3 \rightarrow R_3 - 3R_1$:

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & -4 & 8 \\ 0 & 4 & -8 \end{bmatrix}$$

$R_2 \rightarrow R_2 / -4$:

$$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -2 \\ 0 & 4 & -8 \end{bmatrix}$$

From R_2 : $v_2 - 2v_3 = 0 \Rightarrow v_2 = 2v_3$

From R_1 : $v_1 - v_2 + v_3 = 0 \Rightarrow v_1 = v_2 - v_3 = 2v_3 - v_3 = v_3$

Choose $v_3 = 1$:

Then $v_1 = 1$ and $v_2 = 2$

Third eigenvector: $v_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

Normalize:

$$u_3 = \frac{1}{\sqrt{1+4+1}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Normalized Eigenvectors:

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad u_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad u_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Step 4: Form the Orthogonal Matrix P

The orthogonal matrix P has normalized eigenvectors as columns:

$$P = [u_1 \quad u_2 \quad u_3]$$

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

Verification of orthogonality:

Check $u_1 \cdot u_2 = 0$:

$$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}}(-1)(1) + 0 \cdot \frac{-1}{\sqrt{3}} + \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}}(1)(1) = -\frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}} = 0$$

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Step 5: The Canonical Form

Under the orthogonal transformation $x = Py$:

$$Q = x^T A x = y^T (P^T A P) y = y^T D y$$

where

$$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Canonical Form:

$$Q = -2y_1^2 + 3y_2^2 + 6y_3^2$$

Step 6: Transformation Equations

The relationship between old and new variables:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Explicitly:

$$x_1 = -\frac{y_1}{\sqrt{2}} + \frac{y_2}{\sqrt{3}} + \frac{y_3}{\sqrt{6}}$$

$$x_2 = -\frac{y_2}{\sqrt{3}} + \frac{2y_3}{\sqrt{6}}$$

$$x_3 = \frac{y_1}{\sqrt{2}} + \frac{y_2}{\sqrt{3}} + \frac{y_3}{\sqrt{6}}$$

Step 7: Nature and Properties of the Quadratic Form

Analysis of eigenvalues:

- $\lambda_1 = -2$ (negative)
- $\lambda_2 = 3$ (positive)
- $\lambda_3 = 6$ (positive)

Since the eigenvalues have mixed signs (one negative, two positive), the quadratic form is **INDEFINITE**.

Properties:

- **Index:** $p = 2$ (number of positive eigenvalues)
- **Number of negative eigenvalues:** $q = 1$
- **Signature:** $p - q = 2 - 1 = 1$
- **Rank:** 3 (all eigenvalues are non-zero)
- **Nature:** Indefinite (can take both positive and negative values)

Final Summary

Complete Solution Summary:

1. Original Quadratic Form:

$$Q = x_1^2 + 5x_2^2 + x_3^2 + 2x_1x_2 + 2x_2x_3 + 6x_3x_1$$

2. Matrix Representation:

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

3. Eigenvalues:

$$\lambda_1 = -2, \quad \lambda_2 = 3, \quad \lambda_3 = 6$$

4. Canonical Form:

$$Q = -2y_1^2 + 3y_2^2 + 6y_3^2$$